

How many ‘love languages’ are there, really?



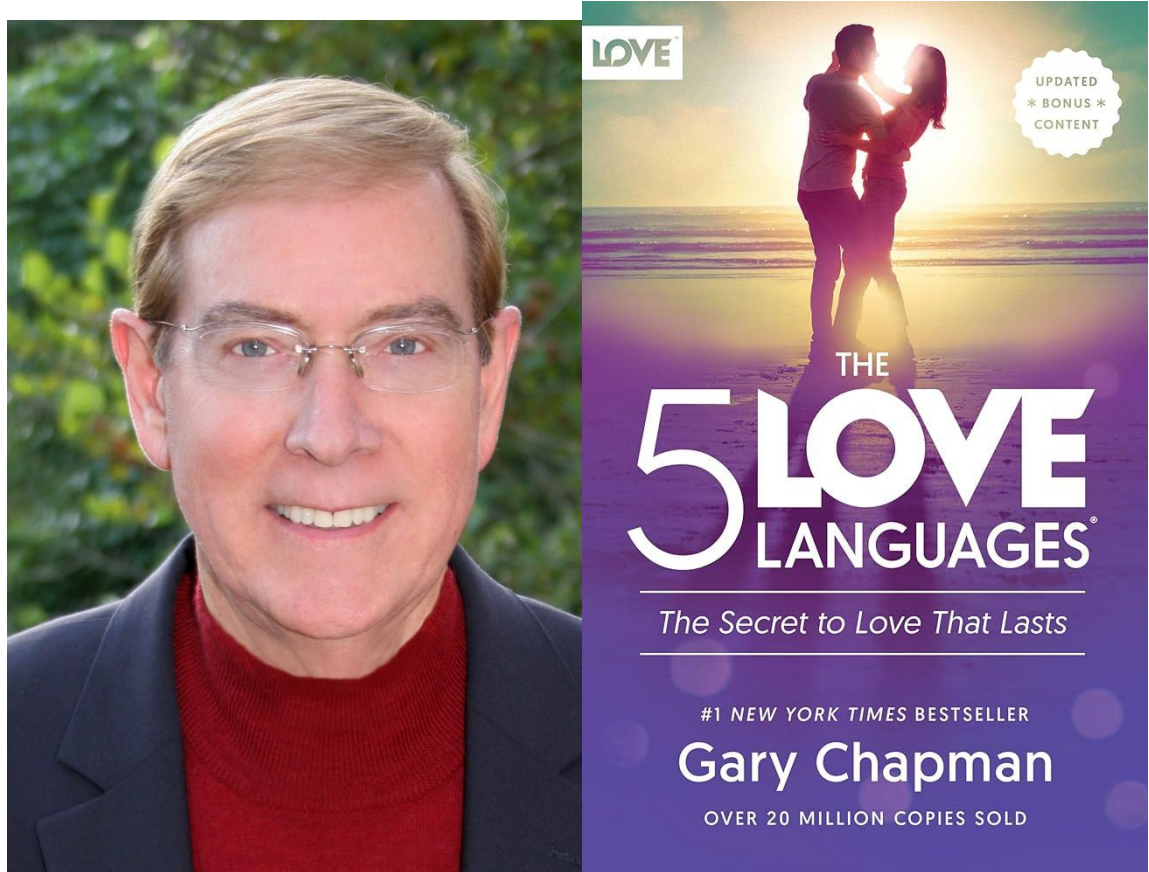
Ryan Dobson¹, David Lewis², and Geoffrey Miller¹

¹University of New Mexico, ²Murdoch University



Background

- Gary Chapman (b. 1938) was a Baptist pastor who counselled many couples
- His book *The 5 Love Languages* (1992) sold over 20 million copies in 49 languages
- He argued that people give and receive love in 5 ways: gifts, quality time, physical touch, acts of service, and words of affirmation
- Empirical research on love languages raises doubts about their factor structure (Impett et al., 2024)
- We wanted to test this factor structure with a larger sample and better survey methods, including separate questions about expressing and receiving love



Online Survey Methods

- Participants completed a short survey (approx. 6-9 minutes) where they reported:
- Basic demographic information (e.g., age, relationship status, sexual orientation)
 - The extent to which they found 29 items (approx. 6 per love language) as 1 = *not at all an expression of their love for me* to 5 = *extremely an expression of their love for me* (i.e., reception)
 - The extent to which they found the same 29 items as 1 = *not at all an expression of my love* to 5 = *extremely an expression of my love* (i.e., expression)

Several data quality checks (e.g., Captcha verification, fast responders [< 4 minutes], and reverse coded questions) were used to filter bad responders.



Demographics

- $N = 740$ (601 recruited from X and 139 university students)
 - 55% male
 - 77% heterosexual
 - $M_{age} = 34.3$
 - $SD_{age} = 12.1$
- Relationship Status:
- 61% exclusive relationship
 - 29% single
 - 10% alternative relationship styles (e.g., polyamorous, swingers)

Factor Analysis

- **An initial confirmatory factor analysis (CFA) specified to match Chapman’s five-factor model did not fit adequately in the reception or expression format.**
- The sample was split into two random halves to perform an exploratory factor analysis (EFA) on half of the sample and a confirmatory factor analysis on the other half of the sample.
- Multiple criteria were used to identify a plausible number of factors (e.g., scree plot inspection and parallel analysis); five factors was determined to be a reasonable solution in both formats.
- CFA on the opposite sample used for EFAs found that full models did not provide adequate fit.
- However, single factor models (with minimal modifications) fit well for each factor in both the expression and reception format and reliability was good for each factor (α range = .79 - .87).

Want to Collaborate?

Ryan Dobson: rdobson@unm.edu

Factor 1: Time and Attention Investment (Quality Time)

Item	R	E
Making a special effort to spend time with me	.81	.77
Making a sacrifice to spend extra time with me	.80	.67
Spending their free time with me	.78	.53
Giving their full attention to me when they see me	.62	.57
Taking time out of their day to support me	.49	.53
Really listening to me	.48	.55
Having a quality conversation	.43	.48

Note. The “R” and “E” columns are pattern matrix factor loadings for the reception and expression solution, respectively. Items displayed are in reception format.

Factor 2: Simple Displays of Affection (Physical Touch and Words of Affirmation)

Item	R	E
Telling me that they love me	.69	.58
Telling me that they miss me	.62	.55
Giving me a kiss	.61	.58
Telling me that they are committed to me	.61	.43
Hugging me	.59	.72
Telling me how important I am to them	.59	.44
Holding my hand	.57	.69
Complimenting my appearance	.35	.43



Factor 3: Thoughtful Acts (Gifts)

Item	R	E
Giving me a surprise present when there’s no special occasion	.85	.87
Randomly getting me a small gift	.79	.79
Giving me a thoughtful birthday gift	.60	.40
Writing sweet notes to me	.45	.50

Factor 4: Pleasing Me Sexually

Item	R	E
Performing sexual acts for me	.88	.90
Having sex with me	.76	.76
Performing oral sex on me	.74	.78

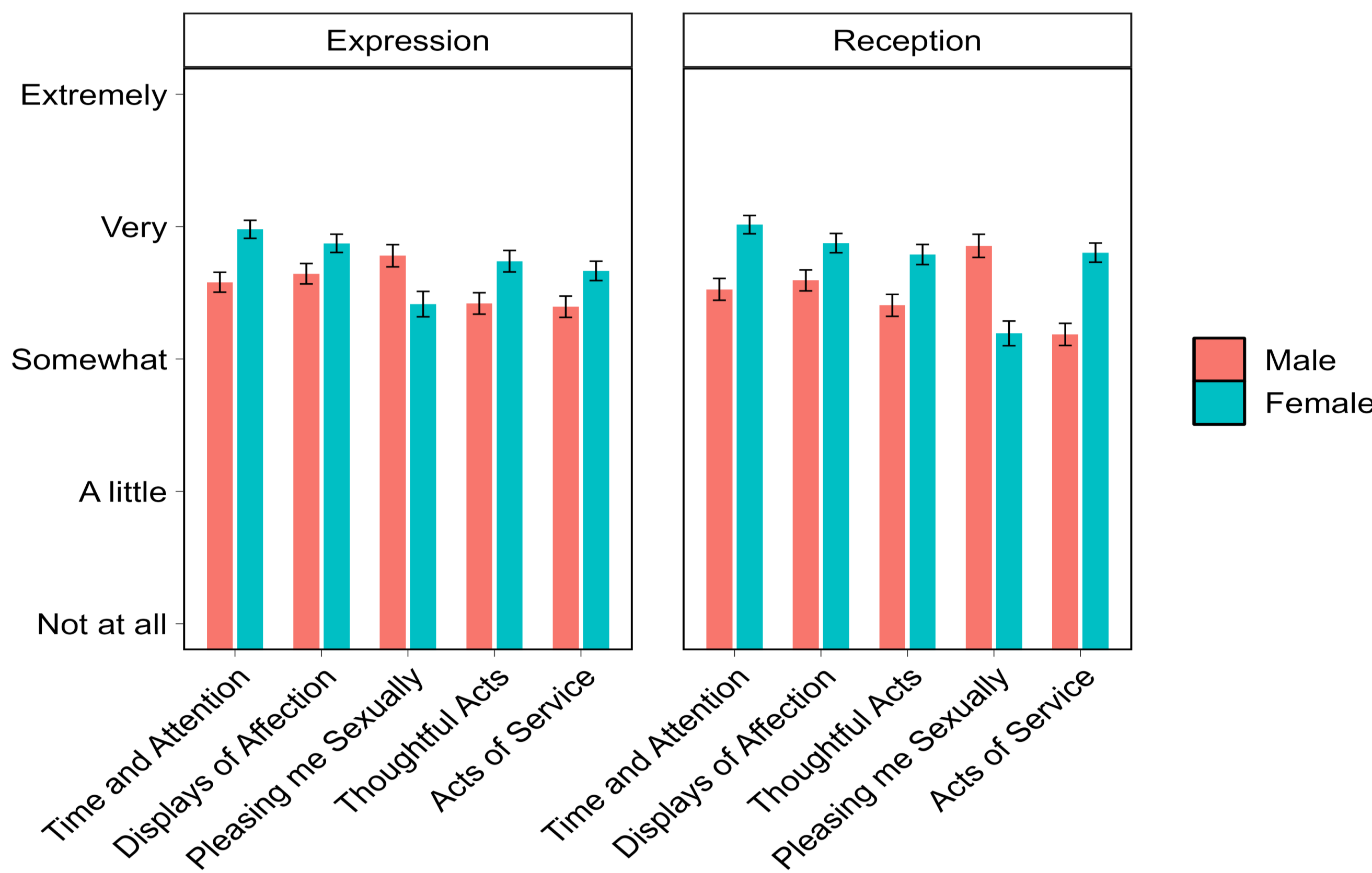
Factor 5: Acts of Service (and financial investment)

Item	R	E
Finishing a chore for me when I don’t have the time to do it	.79	.51
Running errands for me	.63	.52
Helping me keep things cleaned up	.63	.54
Paying for our dates	.29	.47
Paying for my favorite snack	.33	.57

Factors are moderately correlated

Inter-factor correlations from the five-factor reception (upper) and expression (lower) exploratory factor analyses					
	1	2	3	4	5
1. Time and Attention Investment	-	0.47	0.28	0.08	0.38
2. Simples Displays of Affection	0.42	-	0.29	0.18	0.29
3. Thoughtful Acts	0.29	0.28	-	0.07	0.33
4. Pleasing Me Sexually	0.08	0.32	0.15	-	0.05
5. Acts of Service	0.25	0.16	0.22	0.13	-

In both formats, females score higher in every love language other than pleasing me sexually, where males score higher



What was Chapman right about?

- (1) The current factor solution showed some remarkable similarities to Chapman’s:
 - Time and Attention investment includes key elements of Chapman’s “quality time”
 - Thoughtful Acts broadly represents Chapman’s “gifts”
- (2) The factor structure of expression and reception only displayed marginal differences and could be considered the same

What Chapman overlooked

- (1) Chapman’s “words of affirmation and “physical touch” love languages were better represented as a single “simple displays of affection.”
- (2) Explicit sexual acts formed a new “pleasing me sexually” factor, which was highly distinct from all other factors, including simple displays of affection which includes Chapman’s “physical touch.”
- (3) Elements of financial investment were included and were best represented under Chapman’s “acts of service.”

What we overlooked

- (1) Some sub-elements of Chapman’s love languages are not included (e.g., words of encouragement)
- (2) Additional elements—not mentioned by Chapman and not included here—could be incorporated: e.g., “takes interest in my hobbies,” or for couples with children, “helps take care of our child.”